MILITARY SPECIFICATION

NICKEL NITRATE, HEXAHYDRATE, ANALYZED REAGENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one analyzed reagent grade of nickel nitrate, hexahydrate (hereinafter referred to as nickel nitrate).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-W-572	-	Wood-Preservative; Water-Repellent.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Fiberboard.
PPP-C-186	-	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-C-301	-	Chemicals, Dry and Paste, Packaging and Packing of.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

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MIL-P-15011	-	Pallets, Material Handling, Wood, Post Construction, 4-Way Entry.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite "49 CFR 171-179.")

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193-66 - Reagent Water

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 Chemical characteristics. Nickel nitrate shall conform to the chemical characteristics of table I when tested as specified therein.

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Table I. Chemical characteristics

Characteristic	Percent by weight: Minimum	Maximum	Test paragraph
Assay [as $Ni(NO_3)_2 \cdot 6H_2O$]	99.0	-----	4.2.4.1
Insoluble matter	----	0.010	4.2.4.2
Chloride (Cl)	----	0.002	4.2.4.3
Sulfate (SO_4)	----	0.010	4.2.4.4
Alkalies and earths (as SO_4)	----	0.20	4.2.4.5
Cobalt (Co)	----	0.10	4.2.4.6
Copper (Cu)	----	0.010	4.2.4.7
Iron (Fe)	----	0.005	4.2.4.8
Lead (Pb)	----	0.005	4.2.4.9
Zinc (Zn)	----	0.030	4.2.4.10

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the nickel nitrate produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

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4.2.2.2 For test. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 1/4 pound shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

Table II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2-25	2
26-150	3
151-1,200	5
1,201-7,000	8
7,001-20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled and closed unit, intermediate, or shipping container, as applicable, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 4.0 percent defective:

(a) Contents per container not as specified.
(b) Container not as specified.
(c) Container closure not as specified.
(d) Container damaged or leaking.
(e) Fiberboard fillers and pads, strapping, or taping not as specified.
(f) Wood preservation incorrect or missing (level A packing only).
(g) Marking incorrect, missing, or illegible.

4.2.3.2 For test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 1/4 pound. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

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4.2.4.1 Assay [as $Ni(NO_3)_2 \cdot 6H_2O$]. Weigh to the nearest milligram (mg) approximately 2 grams (g) of the specimen and dissolve in water to make 200.0 milliliters (ml) of solution. Transfer a 50.0-ml aliquot to a beaker and dilute to 200 ml with water. Add 1 g of sodium citrate, heat to boiling, add a solution of 0.6 g of sodium dimethylglyoxime in 100 ml of warm alcohol to the boiling solution, add 5 ml of 28-percent ammonium hydroxide solution, and allow to stand overnight. Filter through a tared filter crucible weighed to the nearest 0.1 mg and wash first with hot water and then with 50-percent alcohol. Dry the filter crucible to constant weight at 115° ± 2° C, cool to room temperature in a desiccator, and weigh to the nearest 0.1 mg. Calculate the percent nickel nitrate, hexahydrate as follows:

$$\text{Percent nickel nitrate, hexahydrate} = \frac{402.8\ (A - B)}{W}$$

where: A = Weight of crucible and residue in grams,
B = Weight of crucible in grams, and
W = Weight of specimen in grams.

4.2.4.2 Insoluble matter. Weigh to the nearest milligram approximately 20 g of the specimen and dissolve in 150 ml of water containing 2 ml of concentrated nitric acid. Heat on a steam bath for 1 hour. Filter through a tared filter crucible weighed to the nearest 0.1 mg, wash thoroughly with hot water, dry at 105° to 110° C for 1 hour, cool to room temperature in a desiccator, and weigh to the nearest 0.1 mg. Calculate the percent insoluble matter as follows:

$$\text{Percent insoluble matter} = \frac{100\ (A - B)}{W}$$

where: A = Weight of crucible and residue in grams,
B = Weight of crucible in grams, and
W = Weight of specimen in grams.

4.2.4.3 Chloride. Weigh 2.000 g of the specimen and dissolve in 20 ml of water containing 2 ml of concentrated nitric acid. Divide the solution into two equal portions. To one portion, add 1 ml of 1.7-percent silver nitrate solution, bring to a boil, allow to stand for 10 minutes, cool to room temperature, filter through nitric acid wash filter paper, and retain the filtrate for use as a control. To the second portion of the specimen solution, add 1 ml of 1.7-percent silver nitrate solution. Any turbidity produced shall be no greater than that produced when 0.02 mg of chloride ion is added to the control and the volumes of both solutions are made equal.

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4.2.4.4 Sulfate. Weigh to the nearest milligram approximately 5 g of the specimen and dissolve in 10 ml of water. Add 10 ml of concentrated hydrochloric acid and evaporate to dryness on a steam bath. Take up the residue with 100 ml of water, add 1 ml of concentrated hydrochloric acid, heat to boiling, add 5 ml of 12-percent barium chloride solution, and allow to stand overnight. Filter any precipitate formed through a tared filter crucible weighed to the nearest 0.1 mg, wash with water, ignite, cool to room temperature in desiccator, and weigh to the nearest 0.1 mg. Calculate the percent sulfate as follows:

$$\text{Percent sulfate} = \frac{41\ (A - B)}{W}$$

where: A = Weight of crucible and precipitate in grams,
B = Weight of crucible in grams, and
W = Weight of specimen in grams.

4.2.4.5 Alkalies and earths. Weigh to the nearest milligram approximately 2 g of the specimen and dissolve in 95 ml of water. Add 5 ml of 28-percent ammonium hydroxide solution, precipitate the nickel with hydrogen sulfide gas, and filter. Evaporate 50 ml of the filtrate in a tared evaporating crucible weighed to the nearest 0.1 mg on a steam bath to a volume of 10 ml. Refilter if not clear, evaporate the solution to dryness, ignite, cool to room temperature in a desiccator, and weigh to the nearest 0.1 mg. Calculate the percent alkalies and earths as follows:

$$\text{Percent alkalies and earths} = \frac{200\ (A - B)}{W}$$

where: A = Weight of crucible and residue in grams,
B = Weight of crucible in grams, and
W = Weight of specimen in grams.

4.2.4.6 Cobalt. Weigh 2.000 g of the specimen in 40 ml of water, add 1 ml of 10-percent sodium hydroxide solution and 2 ml of acetic acid and heat to boiling. Add a filtered solution of 5 g of potassium nitrite in 10 ml of water, stir thoroughly, digest on a steam bath for 30 minutes, and allow to stand at room temperature overnight. Filter through a filter crucible and wash with small quantities of cold 5-percent ammonium nitrate solution containing 1 percent of acetic acid until free from nitrite. Place the filter crucible and contents in the same beaker in which the precipitation was made, add 10.0 ml of 0.1N potassium permanganate solution, 25 ml of 10-percent sulfuric acid, and sufficient water to cover the filter crucible. Heat the solution to 50° + 2° C and allow to stand at this temperature until the yellow precipitate has dissolved. Cool to room temperature, add 1 g of potassium iodide crystals and 5 ml of starch

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indicator solution, and titrate the liberated iodine with 0.1N sodium thiosulfate solution. (The liberated iodine represents the excess of permanganate in the solution.) No more than 2.05 ml of permanganate shall be consumed.

4.2.4.7 Copper. Weigh 1.000 g of the specimen and dissolve in 2 ml of concentrated hydrochloric acid and sufficient water to make 50 ml of solution. Add 2 ml of 1-percent mercuric chloride solution and saturate with hydrogen sulfide gas. Filter through a filter crucible, wash with saturated hydrogen sulfide water, and ignite. Dissolve the residue by warming with 0.5 ml of concentrated nitric acid and a few drops of water. Dilute to 10 ml with water and filter if necessary. Add 1 g of ammonium acetate and dissolve in the solution, then add 5 drops of freshly prepared 10-percent potassium ferrocyanide solution. Any red color produced shall be no darker than that produced in a control containing 0.1 mg of copper ions and the same quantities of concentrated nitric acid, ammonium acetate, and potassium ferrocyanide used in the specimen test in the same final volume as the specimen.

4.2.4.8 Iron. Weigh 1.000 g of the specimen and dissolve in 20 ml of water. Add 2 ml of concentrated hydrochloric acid and 5 drops of concentrated nitric acid. Bring the solution to a boil and add sufficient ammonium hydroxide to dissolve the precipitate first formed. Filter and wash with 2.5-percent ammonium hydroxide solution until the washings are colorless. Dissolve any precipitate on the filter with 5 ml of hot 10-percent hydrochloric acid and dilute to 50 ml with water. To a 20-ml aliquot of the solution, add 2 ml of concentrated hydrochloric acid and dilute to 50 ml with water. Add 30 to 50 mg of ammonium persulfate and 3 ml of 25-percent ammonium thiocyanate solution, and mix. Any red color produced shall be no darker than that produced in a control containing 0.02 mg of iron ion, 2 ml of concentrated hydrochloric acid, and the same quantities of ammonium persulfate and ammonium thiocyanate used in the specimen test in the same final volume as the specimen.

4.2.4.9 Lead. Weigh 1.00 g of the specimen and dissolve in 10 ml of water containing 1 g of sodium acetate. Add 5 drops of glacial acetic acid and 1 ml of 1-percent potassium chromate solution. No turbidity shall be produced within 5 minutes.

4.2.4.10 Zinc. Weigh 1.000 g of the specimen and dissolve in 40 ml of water. Heat to boiling and add 20 ml of 10-percent sodium hydroxide solution. Boil for 2 minutes, allow to cool to room temperature, dilute to 80 ml with water, and filter. To 40 ml of the filtrate, add 10 ml of saturated hydrogen sulfide water. Allow to stand for 5 minutes. Any turbidity produced after 5 minutes shall be no greater than that produced in a blank containing 10 ml of 10-percent sodium hydroxide solution, 30 ml of water, and 10 ml of saturated hydrogen sulfide water.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. One-quarter pound + 0.05 ounce of nickel nitrate shall be packaged in accordance with the applicable level A packaging requirements of PPP-C-301 in a type I, class 1 widemouth glass bottle, in inclosure b as specified therein. The glass bottle shall conform to glass type b, light penetration grade optional, with outer seal A band or tape of PPP-C-186.

5.1.1.1 Intermediate packaging. Twelve bottles of nickel nitrate, packaged as specified in 5.1.1, shall be intermediately packaged in accordance with the applicable requirements of PPP-C-301 except that table I therein shall not be used. The bottles shall be arranged in a three bottle by four bottle pattern. Fiberboard separators shall not be required.

5.1.2 Level C. One-quarter pound of nickel nitrate shall be packaged as specified in 5.1.1 except inclosure b shall not be required.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Four intermediate packages, prepared as specified in 5.1.1.1, shall be packed upright in a nailed wood, cleated plywood, or wirebound wood box as specified (see 6.2). The nailed wood box shall conform to class 2, style 4-1/2, for a type 2 average load of PPP-B-621. The cleated plywood box shall conform to type overseas, style I or J, for a type 2 average load of PPP-B-601. The wirebound box shall conform to class 3, style optional, for a type 2 average load of PPP-B-585. The use of fiberboard fillers and pads shall be in accordance with PPP-C-301 to assure a tight pack. Each container shall be closed and strapped using galvanized strapping as specified in the appendix to the applicable box specification. In addition, unless specific provisions are made for preservation in the applicable box specification, each box or the finished, unpainted wood parts thereof shall be completely immersed for a minimum of 3 minutes in a water-repellent wood preservative conforming to composition A of TT-W-572.

5.2.2 Level B. Packing shall be as specified in 5.2.1 except that the box shall conform to style RSC, grade V3c, of PPP-B-636. The box shall be closed in accordance with the requirements for taping style RSC boxes as exterior containers and then reinforced by tape banding as specified in the appendix to PPP-B-636. No preservative shall be used.

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5.2.3 Level C. Twelve bottles of nickel nitrate, packaged as specified in 5.1.2, shall be packed in a fiberboard box conforming to style RSC, type CF, class domestic, variety SW, grade optional of PPP-B-636. The box shall be fitted with fiberboard half-slotted separators which shall form close fitting cells for each of the bottles and which shall be of the same height as the bottles. All inner faces of the box shall be lined with full box-sized fiberboard pads. Additional pads shall be added where needed to assure a tight fit. Fiberboard pads and separators shall conform to type CF, class domestic, variety SW of PPP-F-320. The box shall be closed in accordance with the requirements for taping as specified in the appendix to PPP-B-636.

5.3 Marking. In addition to the special marking for unit containers specified in 5.3.1 and any marking required by the contract or order (see 6.2), unit and intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129 and Department of Transportation (DOT) regulations, including the DOT yellow "Oxidizing Material" label.

5.3.1 Special marking. Each unit container shall be durably and legibly marked to show the following special information.

(a) Empirical formula.
(b) Formula weight.
(c) Manufacturer's lot number.
(d) Assay.
(e) Analysis or maximum limits of impurities.
(f) Precautionary marking:

DANGER

POISON

May be fatal if swallowed.
Avoid breathing dust or spray mist.
Avoid contact with skin, eyes, or clothing.
Wash thoroughly after handling.
If taken internally, get immediate medical attention.

5.4 Palletization. When specified in the contract or order (see 6.2), shipping containers shall be palletized in accordance with the applicable requirements of MIL-STD-147 using load type I except that the strapping shall be restricted to galvanized steel of 1.25-inch minimum width and 0.035-inch minimum thickness. The load tie-down

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straps parallel to the 48-inch pallet dimension shall be modified to serve as load unitizing straps, containing the load but not the pallet. One load tie-down strap shall be added to, and centered between, those straps which are normally required to be parallel to the 40-inch pallet width. One horizontal strap shall be required for each tier of containers. The pallet shall conform to style 1, type optional, class 1, preserved of MIL-P-15011 and all wood storage aids, if used, shall be preserved as specified for the pallet.

6. NOTES

6.1 Intended use. Nickel nitrate is intended for use as a laboratory reagent.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) Kind of box required for level A packing (see 5.2.1).
(d) Additional marking, if required (see 5.3).
(e) Palletization, if required (see 5.4).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

6.5 Supersession data. This specification includes the requirements of Edgewood Arsenal Purchase Description 4-190, dated 25 October 1952.

Custodians:

Army - MU
Air Force - 68

Review activities:

Army - MD, MU(FA)
Air Force - 68
DSA - GS

User activities:

Army - MI, SM

Preparing activity:

Army - MU(EA)

Project No. 6810-0455

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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OFFICIAL BUSINESS

Commanding Officer
Edgewood Arsenal
ATTN: SMUEA-QAES
Edgewood Arsenal, Maryland 21010

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